



DEALER BEST PRACTICE SERIES

Machine Rebuild Using Pack By Dealer Work Order

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

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1.0 Introduction

Purpose: This Best Practice has been developed to provide dealers with a proven machine rebuild process that utilizes the Pack By Dealer Work Order Program (PBWO) available through Caterpillar, Inc. In addition, it offers suggestions on how to improve rebuild operations in order to deliver efficiencies with dealer parts and service operations, which ultimately improve our customer's machine availability and productivity. Note, the following information is specific to the example provided below. Results may vary based on different circumstances.

Process: Machine rebuild programs and processes can vary from one dealer to another and provide customers with different results. Empire Machinery, working on behalf of a high profile customer, was able to develop and implement a machine rebuild program for 793B trucks that consistently meet their customer's expectations of quality, turnaround time, and performance. This process was refined as 61 trucks were rebuilt over the course of 3 years.



Currently, there is limited understanding of the PBWO Program and how it can impact a dealer's rebuild process, parts operations, and service operations. The PBWO is a special service provided by Caterpillar that enables dealers to place orders for machine, engine, or major component rebuilds by the dealer's work order segment. Orders are then packed complete and accurate per the dealer's specifications and shipped when the dealer needs the rebuild parts. An area on each case label is customizable by the dealer and can include such items as the work order segment, operation number, and service manual page numbers. This program was a critical element of the truck rebuild program and enabled Empire to deliver all 61 truck rebuilds on time.

Payoff: The programs and processes described in this Best Practice lead to tangible benefits to both customers and dealers. Some of the potential savings and efficiencies realized include:

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Parts Operations Efficiencies:

- Reductions in time for
 - o Identifying parts
 - o Ordering parts
 - o Expediting parts
 - o Waiting for parts
 - o Parts stocking
 - o Parts picking
 - o Parts verification
 - o Parts returns
- Parts capacity improvements
- Reduction in transportation costs
- No emergency fees
- Inventory carrying cost reductions
- Billing efficiency improvements

Service Operations Efficiencies:

- Rebuild efficiency improvements
- Increased workshop capacity
- Technician productivity improvements
- Turnaround time improvements
- Work in process reductions

Customer Related Benefits:

- Availability improvements
- Productivity improvements
- Mean time between repair events improvements
- Cycle time improvements



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2.0 Best Practice Description

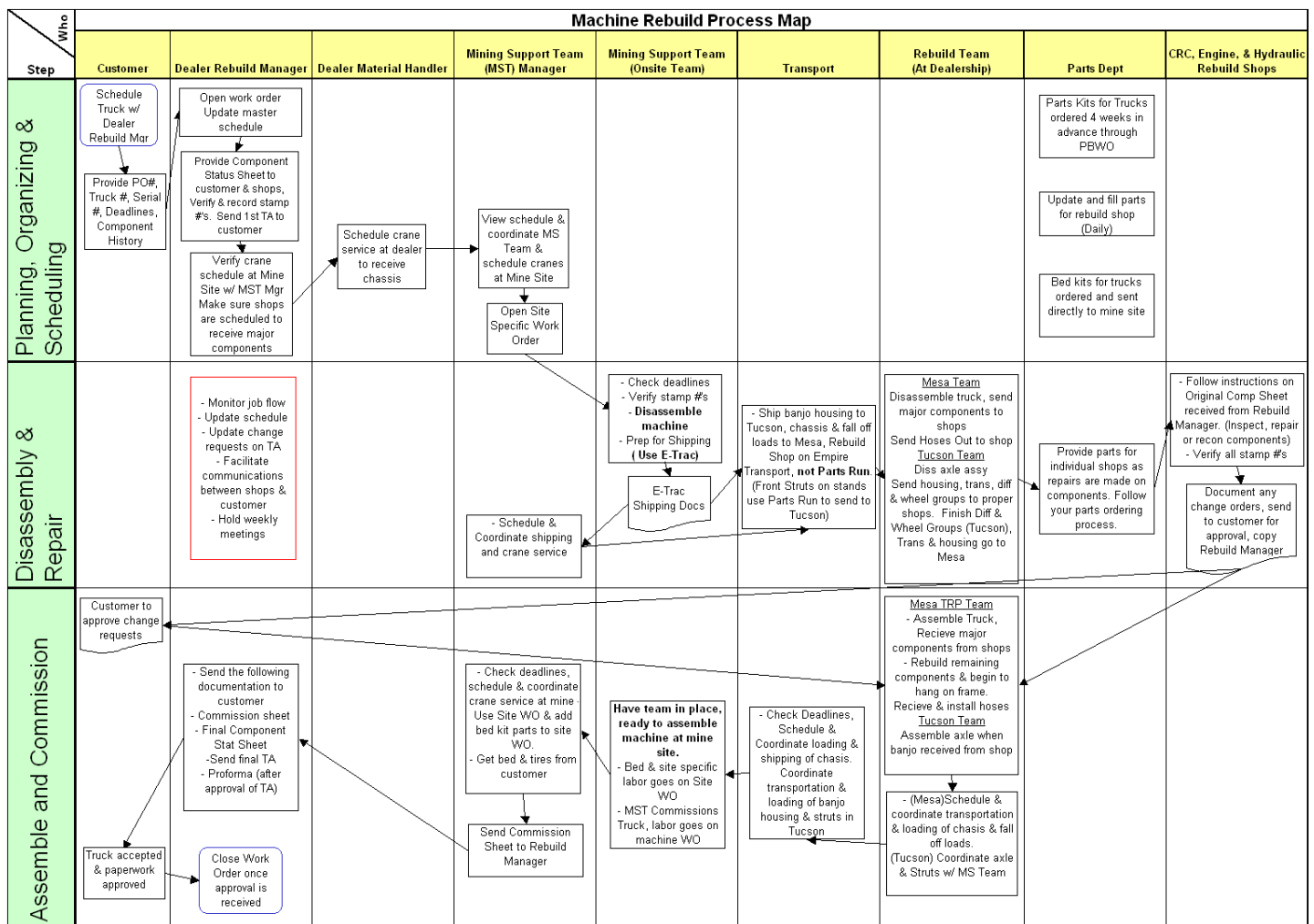
The following machine rebuild Best Practice was developed and modified as 61 trucks were rebuilt at Empire Machinery. This Best Practice focuses on improvements in processes; utilization of the PBWO Program; establishing team focus; communications; training; and parts & service operations. Lessons learned in each area are discussed.

Improved Processes

1) Rebuild Process Map

In order to drive continuous improvements with the rebuild program and meet customer expectations & deadlines, a truck rebuild process map was created. This process map identified all the key players in the rebuild program and outlined specific roles and responsibilities. It further aided the Dealer Rebuild Manager in establishing clear communications across the rebuild teams.

The rebuild process was broken down into 3 main steps: 1) planning, organizing, and scheduling; 2) disassembly & repair; and 3) assemble & commission. Each step in the process was clearly described and an accountable owner was identified.



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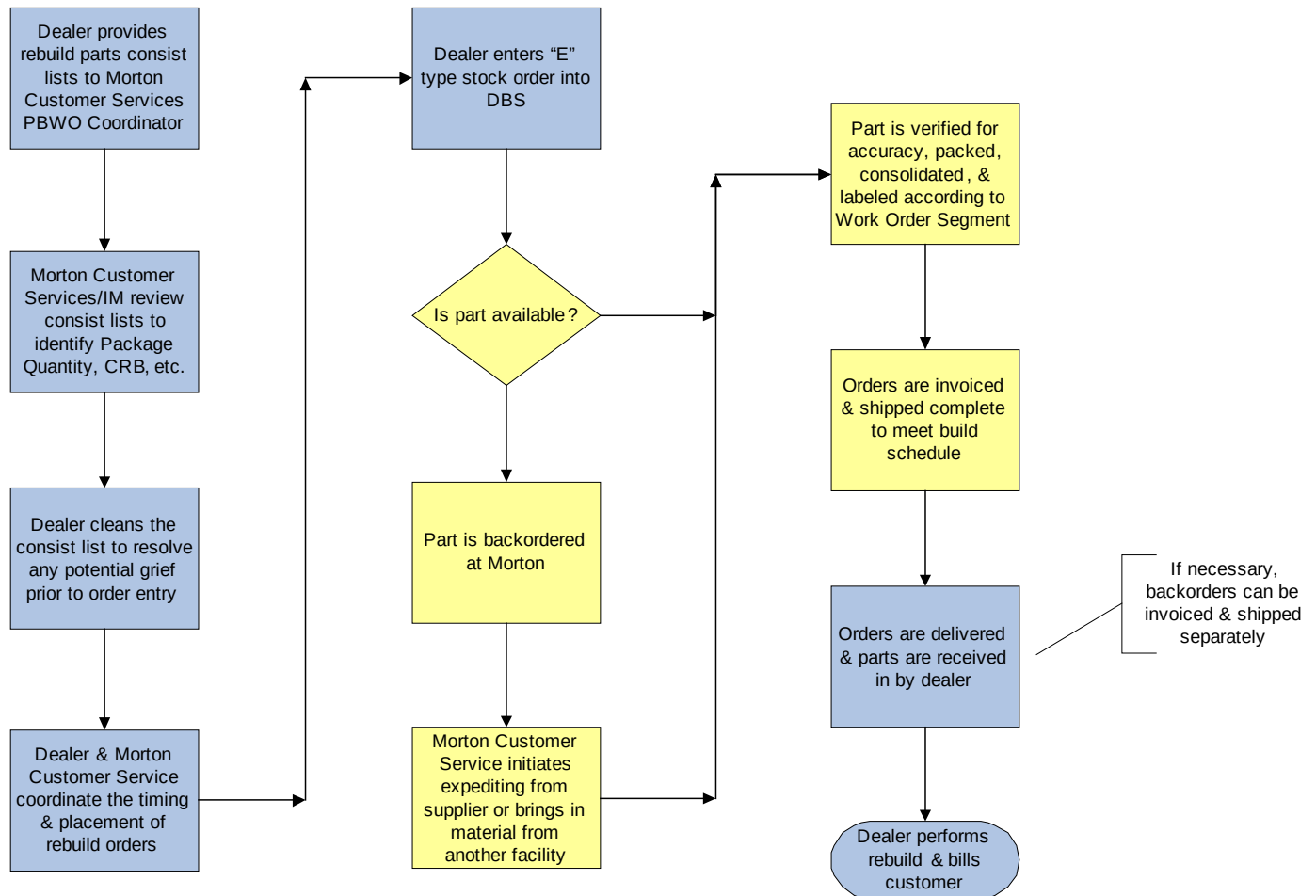
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2) PBWO Program Process

An integral part of the rebuild program was the utilization of the PBWO Program offered by Caterpillar. In order to use this program the dealer established their work order segments and created parts consists per segment. These segments and parts consists, along with the machine model and serial number, were provided to Customer Services in Morton approximately 60 days prior to the rebuild. Customer Services reviewed the lists to identify package quantities, cancel & replacements, etc. and worked with the dealer to resolve any potential order entry grief. The dealer then placed the order by entering an "E" type stock order into DBS approximately 30 days prior to requiring the parts.

Once the order was received, Caterpillar packed the parts, verified for accuracy, consolidated and labeled the cartons according to the dealer's instructions on the work order segment. Customer Services initiated expediting from suppliers or brought material in from another facility if parts are not available in Morton. One consolidated invoice was generated for the parts orders and they were shipped freight prepaid (for dealers in Americas North or to the port of exit for all other dealers). The following process map provides a high level overview.



Parts Kits

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A multi-tabbed Excel spreadsheet was created that listed parts kits/consists for each of the work order segments. These were used to later place orders.

Example of Parts Kit ordered through the PBWO Program

793BRAKELN order to seg 22 793 BRAKE LINE 8X0236			
QTY	Part #	Description	
1	8D9947	CLIP	Included in Kit
1	8X6211	PLATE	
2	5P8112	GROMMET	
4	8X6210	CLAMP	
1	7G3832	GROMMET	
1	1275588	COUPLING	
4	5P7854	CLIP	
2	5P7463	CLIP	
1	1P3703	SEAL	Pulled From Package Quantities
12	6V9746	SEAL	
5	3K0360	SEAL	
14	3D2824	SEAL	
2	2M9780	SEAL	
1	4H6112	BREATHER	
1	6D4246	CLIP	
10	3J1907	SEAL	
12	3K0360	SEAL	
4	3D2824	SEAL	
4	7J9108	SEAL	
2	2M9780	SEAL	
5	6V9746	SEAL	
1	1P3704	SEAL	

Team Focus

One of the key success factors relating to this rebuild program was the dedication and teamwork from all participants. The first step was focused on creating a partnership with the customer where his requirements and timelines were well communicated. In order to support this effort, weekly rebuild meetings were established where all functions of the rebuild program were represented. These included the customer, onsite mining support team (dealer), the rebuild team located at the dealership, the CRC shop, Hydraulic shop, Engine Spec shop, Parts Department, and Wheel Group shop.

The team developed more than 60 parts kits that were ordered through the PBWO program. They coordinated work between the mine site and dealers' rebuild facilities. Roles and responsibilities were defined in order to minimize ambiguity and clarify ownership. Every player knew their role and accepted the responsibility to meet the customer's deadlines.

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Communications

Communication! Communication! Communication! This program would not have been successful without the concerted effort from every rebuild participant to discuss the status of their work, any issues or concerns, and the vision, goal, and expectations of the rebuild program. All technicians had a clear understanding of their roles and responsibilities.

As a result of the weekly team meetings, no issue or problem was ever more than a week old. All issues were coordinated and handled as they occurred.

The truck rebuild schedule was managed by the Rebuild Manager and based upon inputs from the customers. This schedule drove all work. Key elements of the rebuild schedule are listed as follows:

- PO #
- Truck #
- Serial #
- Original mine site
- Destination mine site
- WO #
- Rebuild #
- Frame type (new or rebuild)
- Parts kits ordered
- Location of rebuild (which dealer branch)
- Actual date of disassembly
- Date shipped to dealer
- Date of disassembly at dealer
- Start date of assembly at dealer
- Frame needed by
- Steering cylinders/center steering arms/accumulators needed by
- Hoses needed by
- All pumps needed by
- Engines & TC needed by
- Hoist cylinders & rear struts needed by
- Axle housing, transmissions, differentials needed by
- Date shipped to mine site
- Start date of assembly
- Date of commission
- # days to build
- First rebuild quote (TA)
- Total cost of rebuild (at Customer price level)

Training

Another key success factor relating to the rebuild program concerned the cross training of all technicians. The technicians located at the dealer's rebuild facilities were cross trained with the

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mining support team (onsite team) to see what processes could be improved with each others work. Teams of technicians were formed to complete rebuilds and quality control checks were performed by the other technician teams on all outgoing rebuilds.

Parts & Service Operations Improvements

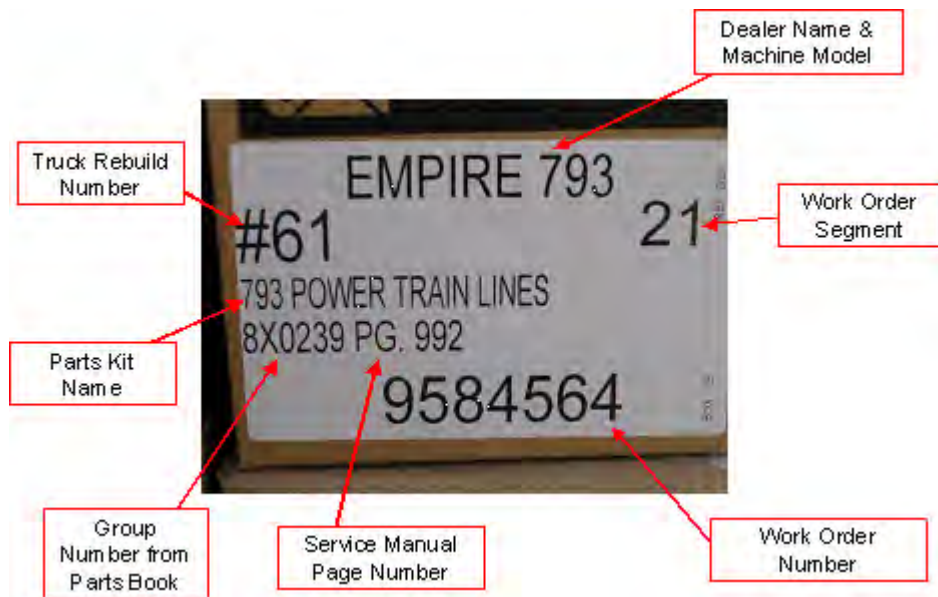
Several parts & service operations improvements were implemented for this rebuild program. Every work order was opened prior to rebuilding each truck so parts kits could be ordered. A template was used for all rebuilds so that the work order segments were always the same from one rebuild to another.

The PBWO Program allowed parts to be shipped from Morton to the workshop bays directly. Each rebuild team had the responsibility of organizing the parts kits into their rebuild bay. The parts kits were never stocked or picked from the dealer’s normal parts inventory.



Each carton was labeled per the dealer’s specifications. In this example, the label contained the dealer name, machine model, truck rebuild number, parts kit name, group number, service manual page number, work order number, and work order segment.

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All packaged quantities were placed in Vidmars and utilized in the workshop bays so technicians were not waiting on parts. Transmission rebuild kits were developed and placed in wheeled carts.



The Parts Department maintained wheeled rebuild kits that were utilized by the engine spec shop.

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As hoses were removed from the truck, each was labeled and placed in the appropriate colored bin and then sent to the hydraulic shop. The hoses were either refurbished or replaced with new and then returned to the rebuild area in the bins. White bins were for cooling system hoses. Tan bins were for hoist and hydraulic system hoses. Grey bins were for brake system hoses. Green bins were for transmission and powertrain systems. Red bins were for steering system hoses and blue bins were grease system hoses.



Complete new cab assemblies were utilized in the rebuild program to improve turnaround time, technician productivity and provided a lower cost option for the customer vs. rebuilding the old cabs.

There are two options available for complete cab assemblies available through the Caterpillar parts system:

Upgrade Cabs – These cabs are very similar to current production cabs, including current markings and paint schemes. Cabs include the latest engineering upgrades and control modules.

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These cabs are available for Large Off-Highway trucks and are typically available in the inventory system.

Replacement Cabs – These cabs are serial number specific and custom-built to match the original consist of the machine when the machine was new. These cabs are not placed in inventory, but are made as ordered. Lead times can vary.

For more information on Upgrade/Replacement cabs, please see Information Release Memo PELJ0501.



3.0 Implementation Steps

The rebuild process and associated Best Practices were developed over the course of 3 years; however, they can quickly be replicated and incorporated with other dealer rebuild programs. Even though this program was developed as a result of rebuilding 61 - 793B trucks, it can be utilized with a much smaller number of machines. Several of the processes can be implemented with existing dealer rebuild programs or modified/adapted to meet specific dealer needs.

The PBWO Program offered by Caterpillar is not only appropriate for dealer rebuilds; but also for Certified Rebuilds, Certified Powertrains, Emissions Upgrades and major component rebuilds. Again, a large number of machine rebuilds are not required to ensure this program is cost effective.

Additional Rebuild steps included:

Frames

Hybrid versions of the 793D frame were used in the rebuild program. These hybrid frames are available through Decatur, and are designed to be compatible with the 793B and 793C models.

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The hybrid frames are further retrofitted by the rebuilding dealer to become a complete B or C frame. Full retrofit instructions can be found in Special Instruction REHS3396. The following pictures include examples of modifications made by the dealer to complete the hybrid frame conversion.



Brackets not found on the “D” frames were added including the radiator supports. Jig fixtures were developed to aid in locating the new parts.



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All catwalks, handrails, and engine enclosures were repaired. The platform was fitted on the frame.



Tanks

Fuel tanks, hoist tanks, and steering tanks were inspected and repaired.



Wheel Groups

The rear axle group, which includes both rear wheel stations, differential, transmission, all valves, filters, and line groups were shipped to CRC. The group was then disassembled and rebuilt per the program guidelines.



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All valves and housings were resealed. All hoses and sensors were replaced and all new mounting hardware was used. All components were then reassembled and shipped back to the mine site to meet the chassis.

Axle Housing

The axle housing rebuild process included glass beading the entire housing inside and out. The housing was Magna-fluxed to identify cracks. A dimensional inspection was conducted to determine the extent of wear, warping, and roundness. Gusset plates were added to bottom of the housing for extra strength. All bolt holes were plug welded. All faces, bores, and registers were welded. The housing was Meta-Lax stress relieved. All bores, faces, registers and bolt holes were re-machined to specification. Cleaned and painted the unit inside and out. Built up bottom strut bores if required.



Mining Support Team

The Mining Support Team disassembly duties included removal of tires, fluids, and the bed. Each truck was disassembled in 5 days to 3 transportable loads: chassis, final drive, and the falloff load.



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The Mining Support Team's build team consisted of 4 technicians. Three loads hit the assembly site. It took 7 days from lay out of the loads to the truck being in the dirt.



4.0 Benefits

Dealer Perspective

The dealer realized several benefits/savings from integrating parts and service operations improvements along with utilization of the PBWO program. These benefits were validated through discussions and documentation provided by Jeremy Hefner and Tom Lamb at Empire Machinery.

Benefits relating from parts efficiencies were estimated to be approximately \$58,880. per truck and were broken down as follows:

- Parts Identification - \$43,890. Savings
 - o Prior to the program, technicians were spending 3-4 hours per night at an overtime rate to identify the rebuild parts.
- Parts Ordering - \$612. Savings
 - o Prior to the PBWO Program, it would take the technicians anywhere from 30 minutes to 2 hours to order parts for 1 work order segment. There were 35 work order segments for each truck rebuild.
- Emergency Fees - \$408. Savings
 - o With the PBWO Program, the dealer does not pay any emergency fees.
- Time Spent Waiting for Parts - \$575. Savings
 - o Technicians would typically wait anywhere from 30 minutes to 8 hours for parts depending on how they ordered them
- Parts Stocking - \$5,280. Savings
 - o Parts for the rebuild program were not stocked in the dealer's parts inventory. They went directly to the dealer's rebuild bay.
- Parts Picking - \$256. Savings

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- o Since the rebuild parts were never stock in the dealer's inventory, there was no cost associated with pulling parts when needed by the technicians.
- Parts Returns - \$320 Savings
 - o On initial rebuilds, 30 to 40 parts were returned per rebuild. With the PBWO program, parts returns were reduced to 5 to 10 parts.
- Freight - \$3,054 Savings
 - o With the PBWO Program, the freight was prepaid for Americas North dealers.
- Inventory Carrying Costs - \$4,485. Savings
 - o Since the rebuild parts were never carried in the dealer's inventory, carrying costs were eliminated.

Benefits relating from service operations efficiencies were estimated at \$45,029. per truck and were broken down as follows:

- Rebuild Efficiency - \$33,120 Savings
 - o The rebuild portion of the program (time in shop) went from 10 weeks to 4 weeks, which resulted in reduced labor hours at the current wage rate.
- Technician Productivity - \$5,520. Savings
 - o This metric went from the dealer average of 85% to 95% under the rebuild program, which meant a technician was 6 minutes more productive per rebuild hour. This reduced lost time resulted from the implementation of the best practices previously mentioned.
- Turnaround time - \$6,389. Savings
 - o Turnaround time went from 110 days to 52 days and the financial impact of this reduction in work in process provided this savings.

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**Chassis Being Assembled****Rebuild Bays**

Customer Perspective

The rebuild program utilizing PBWO provided significant benefits to the customer. These included:

- Truck availability went from an average of 85% to 92%.
- Gained 60 days production per truck, which basically paid for the rebuild.
- Mine realized an overall 7% availability improvement the first year of the rebuild, which equated to an extra “truck’s” worth of production.
- Rebuilt trucks incorporated all the latest updates.

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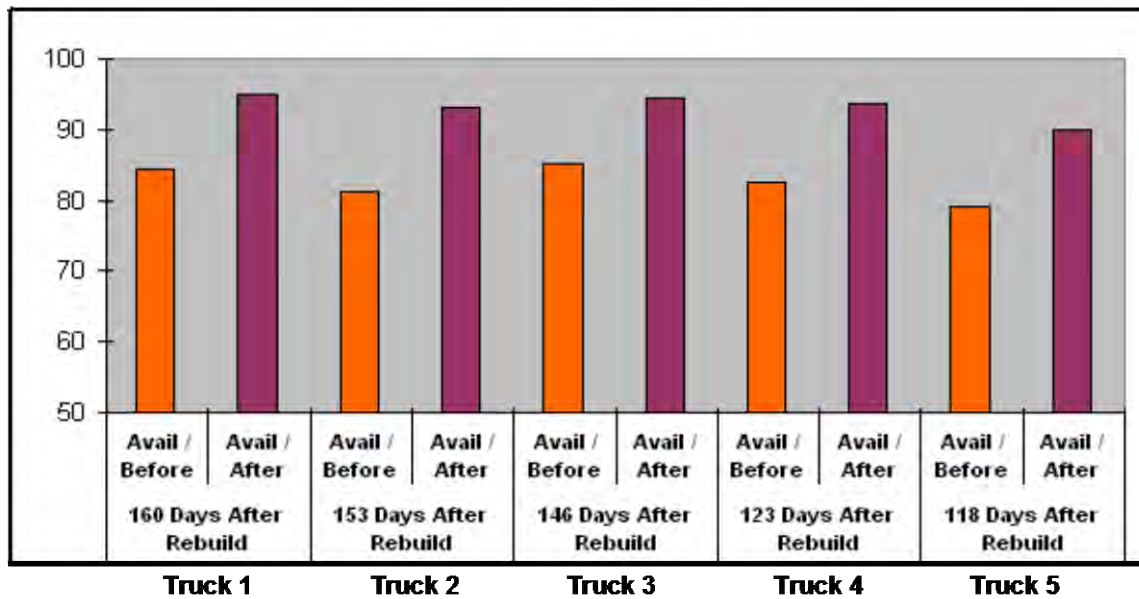
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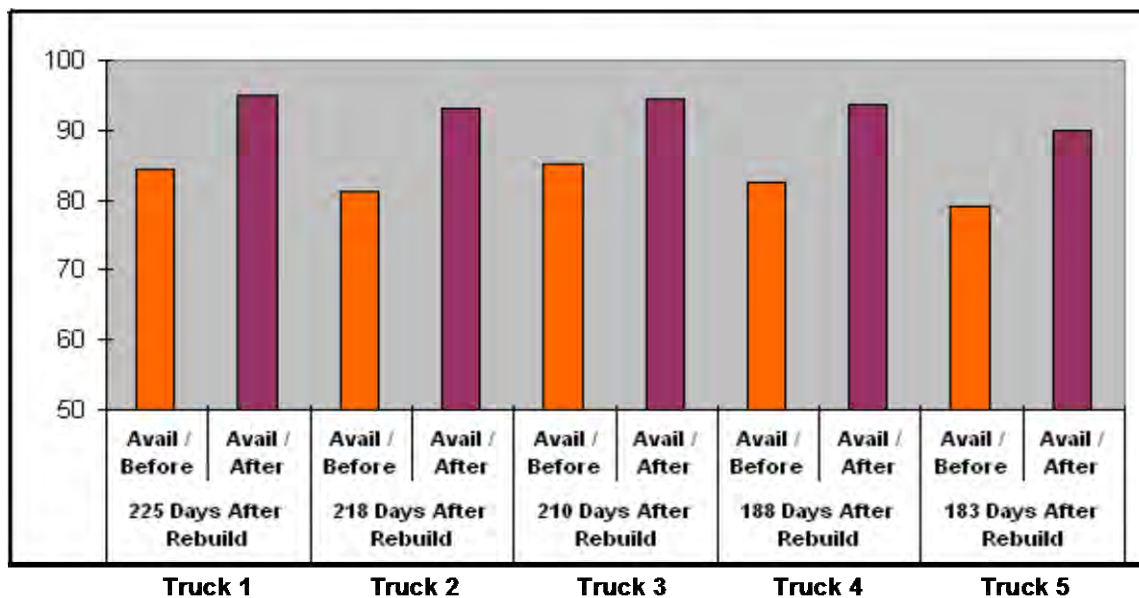
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Customer Success Measured...

Truck Availability

Customer Success Sustained...

Truck Availability

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5.0 Resources Required

With regards to people, the truck rebuild team was comprised of the customer, an overall Rebuild Manager, Material Handler, Mining Support Team Manager (responsible for rebuild operations on the mine site), Mining Support Team (located at the mine site), Transport person, Rebuild Teams, Parts Analyst, and the CRC, Engine, Wheel Group, and Hydraulic shops. See the Rebuild Process Map in Item 2 - Best Practice Description for details regarding the specific roles and responsibilities.

From a cost standpoint, a 5% fee of the total parts order (at Dealer Net) was charged for the specialized services provided by Caterpillar Logistics.

6.0 Supporting Attachments / References

The following documents are available to support the use of the machine rebuild program utilizing the PBWO Program.

- Machine Rebuild Schedule Template (Excel)
- Rebuild Process Map (Excel)
- Component Status Sheet Template (Excel)
- Customer % of Life Component Chart (Excel)
- PBWO Overview Presentation (Powerpoint)
- PBWO Frequently Asked Questions (Word)
- PBWO Dealer Order Entry Presentation (Powerpoint)

7.0 Related Best Practices

8.0 Acknowledgements

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